

# Experiment - 05

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(A)

A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the confirmation rate, which is defined as:

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

If a user has never received any confirmation request, their confirmation rate should be considered 0.

Display the user ID along with the confirmation rate rounded to 2 decimal places. The result can be returned in any order.

**Confirmations**

| user_id | time_stamp          | action    |
|---------|---------------------|-----------|
| 3       | 2021-01-06 03:30:46 | timeout   |
| 3       | 2021-07-14 14:00:00 | timeout   |
| 7       | 2021-06-12 11:57:29 | confirmed |
| 7       | 2021-06-13 12:58:28 | confirmed |
| 7       | 2021-06-14 13:59:27 | confirmed |
| 2       | 2021-01-22 00:00:00 | confirmed |
| 2       | 2021-02-28 23:59:59 | timeout   |

**Signups**

| user_id | time_stamp          |
|---------|---------------------|
| 3       | 2020-03-21 10:16:13 |
| 7       | 2020-01-04 13:57:59 |
| 2       | 2020-07-29 23:09:44 |
| 6       | 2020-12-09 10:39:37 |

Output

| user_id | confirmation_rate |
|---------|-------------------|
| 2       | 0.50              |
| 3       | 0.00              |
| 6       | 0.00              |
| 7       | 1.00              |

### Solution:

```
SELECT S.USER_ID, ROUND(AVG(CASE WHEN ACTION='confirmed' THEN
1 ELSE 0 END), 2) AS CONFIRMATION_RATE
FROM SIGNUPS AS S
LEFT JOIN
CONFIRMATIONS AS C
ON S.USER_ID=C.USER_ID
GROUP BY S.USER_ID;
```

### (B)

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between 2013-10-01 and 2013-10-03, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

We need to calculate:

$$\text{Cancellation rate} = \frac{\text{Cancelled Trips}}{\text{Total Valid Trips}}$$

Only consider trips where:

Client is NOT banned

AND

Driver is NOT banned

and

Date between

2013-10-01

And 2013-10-03

**Users**

| users_id | banned | role   |
|----------|--------|--------|
| 1        | No     | client |
| 2        | Yes    | client |
| 3        | No     | client |
| 4        | No     | client |
| 10       | No     | driver |
| 11       | No     | driver |
| 12       | No     | driver |
| 13       | No     | driver |

**Trips**

| id | client_id | driver_id | city_id | status              | request_at |
|----|-----------|-----------|---------|---------------------|------------|
| 1  | 1         | 10        | 1       | completed           | 2013-10-01 |
| 2  | 2         | 11        | 1       | cancelled_by_driver | 2013-10-01 |
| 3  | 3         | 12        | 6       | completed           | 2013-10-01 |
| 4  | 4         | 13        | 6       | cancelled_by_client | 2013-10-01 |
| 5  | 1         | 10        | 1       | completed           | 2013-10-02 |
| 6  | 2         | 11        | 6       | completed           | 2013-10-02 |
| 7  | 3         | 12        | 6       | completed           | 2013-10-02 |
| 8  | 2         | 12        | 12      | completed           | 2013-10-03 |
| 9  | 3         | 10        | 12      | completed           | 2013-10-03 |
| 10 | 4         | 13        | 12      | cancelled_by_driver | 2013-10-03 |

**Output**

| Day        | Cancellation Rate |
|------------|-------------------|
| 2013-10-01 | 0.33              |
| 2013-10-02 | 0.00              |
| 2013-10-03 | 0.50              |

### Solution:

```

SELECT T.REQUEST_AT AS DAY, ROUND(AVG(CASE WHEN T.STATUS
IN ('cancelled_by_driver', 'cancelled_by_client') THEN 1 ELSE 0 END), 2)
AS CANCELLATION_RATE
FROM
TRIPS AS T
JOIN USERS AS U
ON T.CLIENT_ID = U.USERS_ID
JOIN USERS AS D
ON T.DRIVER_ID=D.USERS_ID
WHERE U.BANNED='No' AND D.BANNED='No' AND T.REQUEST_AT
BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY REQUEST_AT;

```